

Prime Palindrome Demonstration

Title: Prime Palindrome Checker

Objective/Aim:

To write a Java program to check whether a given number is both prime and palindrome.

System Requirements :

Hardware

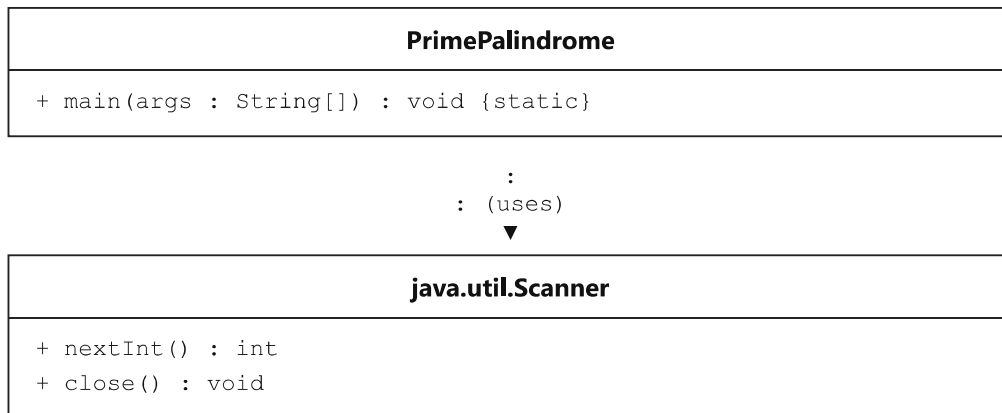
- Standard PC/Laptop
- Intel Core i3/i5 processor or equivalent
- Minimum 4 GB RAM
- Minimum 500 MB free storage
- Keyboard and Monitor

Software

- Operating System: Windows/Linux/macOS
- Java Development Kit (JDK 17 or JDK 21)
- Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
- Command Prompt/Terminal

Algorithm / Logic Description:

1. Import the Scanner class from `java.util` package to read user input.
2. Create a class named `PrimePalindrome` containing the `main()` method.
3. Create a Scanner object and prompt the user to enter an integer.
4. Initialize a variable `count = 0` to keep track of divisors.
5. Execute a loop from 1 to `num`, incrementing `count` each time `num % i == 0`.
6. Reverse the input number using a `while` loop by extracting the last digit (`rem = temp % 10`) and appending it (`rev = rev * 10 + rem`).
7. Check if both conditions are satisfied:
 - The number is prime: `count == 2`
 - The number is a palindrome: `num == rev`
8. If both conditions evaluate to true, print "**Prime Palindrome**"; otherwise, print "**Not a Prime Palindrome**".
9. Close the Scanner object.
10. Stop the program.

UML class Diagram

Relationship: The PrimePalindrome class uses the Scanner class to accept numerical input from the console.

Source Code:

```
import java.util.Scanner;

public class PrimePalindrome {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        // Check for prime number
        int count = 0;
        for (int i = 1; i <= num; i++) {
            if (num % i == 0) {
                count++;
            }
        }

        // Reverse the number to check for palindrome
        int temp = num;
        int rev = 0;
        while (temp > 0) {
            int rem = temp % 10;
            rev = rev * 10 + rem;
            temp = temp / 10;
        }

        // Validate both conditions
        if (count == 2 && num == rev) {
            System.out.println("Prime Palindrome");
        } else {
            System.out.println("Not a Prime Palindrome");
        }

        sc.close();
    }
}
```

```
}  
}
```

Compilation & Execution Commands:

Save file as: PrimePalindrome.java

Compilation: javac PrimePalindrome.java

Execution: java PrimePalindrome

Output Screenshot:

```
PS C:\Users\mmeen\RAVI> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' ... 'PrimePalindrome'  
Enter a number: 23  
Not a Prime Palindrome  
PS C:\Users\mmeen\RAVI>
```

Result:

The Prime Palindrome program was successfully implemented in Java. The program demonstrates reading user input using the Scanner class, checking prime divisibility using iteration, verifying palindrome equality through numerical reversal, and outputting the corresponding status.